

Lecture 19

STAT 109: Introductory Biostatistics

Lecture 19: Confidence Intervals and the Importance of Sampling

Learning outcomes

- Discuss the potential for and impact of **bias** given a description of a sampling method.
 - Identify ways to reduce **bias** and increase **precision** when sampling.
 - Identify a **sampling frame** for a given population.
 - Carry out **simple random sampling**.
 - Classify a sampling method as **convenience**, **simple random**, **cluster**, **systematic**, **stratified**, or **multi-stage**.
-

What is a confidence interval?

A **confidence interval (CI)** provides a range of plausible values for a population parameter based on a sample from the population. It quantifies the uncertainty in our sample estimate. For a 95% CI, we use the approximate formula:

Statistic $\pm 2 \times$ (standard error of the statistic).

Whether that interval is actually *reasonable* for the population we care about depends on **how the sample was obtained** — that is, on the **sampling method**.

Motivation: How many business majors?

Suppose we want to estimate the **proportion of all undergraduate students on campus** who are business majors.

We decide to survey the students in **this Stat 109 class**.

- **In class:** How many students in the room are majoring in business? (Count; call that x . The class size is n .)
- **Example values:** Suppose there are $n = 25$ students and $x = 8$ report being business majors.

Sample proportion: $\hat{p} = \frac{x}{n} = \frac{8}{25} = 0.32$ (32%).

95% confidence interval for p (Wald, assuming conditions hold):

- $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.32 \times 0.68}{25}} \approx 0.093$.
- **95% CI:** $\hat{p} \pm 2 \times SE(\hat{p}) = 0.32 \pm 0.186$, so about **[0.13, 0.51]** or **[13%, 51%]**.

So from the formula we get: *We are 95% confident that the true proportion of all undergraduates who are business majors is between about 13% and 51%.*

Is this interval reasonable?

Critical question: Is it reasonable to treat this interval as applying to **all students on campus**?

- **Population of interest:** All undergraduate students at the university.

- **Sample:** Only students in this Stat 109 class.

Problem: Business majors are **not required** to take Stat 109. So:

- The students in this class are not a random sample of all undergraduates.
- We have a **convenience sample** (whoever is in the room). The proportion of business majors in Stat 109 might be higher or lower than on the whole campus — we have no guarantee the sample is representative.
- So the **confidence interval from the formula** may be misleading: it describes uncertainty due to *random sampling variation in using $\hat{p}$ to estimate p* , but it does **not** account for **bias** due to how we chose the sample. If the sample is biased, the interval may not be a plausible range for the *campus* proportion at all.

Takeaway: The way we **sample** matters. To make confidence intervals that we can trust for the population we care about, we need to think about **sampling methods** and **bias**.

Sampling frame

The **sampling frame** is the list we use to identify and select units from the population. Ideally, the frame matches the population of interest (every member has a chance to be selected; no one is in the frame who isn't in the population, and vice versa). In the business-major example, “students in this Stat 109 class” is a poor frame for “all undergraduates on campus”.

Sampling methods (overview)

We can **classify** how a sample was obtained:

Method	Brief description
Convenience	Select whoever is easy to reach (e.g. this class). Often biased.
Simple random (SRS)	Every subset of n units has the same chance of being selected.
Stratified	Divide population into strata (groups); take a random sample from each stratum.
Cluster	Divide population into clusters; randomly select clusters and often measure all (or a sample of) units in those clusters.
Systematic	From a list, take every k th unit (e.g. every 10th student).
Multi-stage	Combine methods (e.g. randomly select clusters, then randomly select units within clusters).

Lecture 20 will go into more detail on these and on **carrying out** simple random sampling.

Bias and precision

- **Bias:** Systematic tendency to over- or under-estimate the parameter. Comes from *how* we select or measure (e.g. who is in the frame, who responds).
- **Precision:** How similar our estimates would be if we repeated the process (related to standard error and width of the CI). Affected by sample size and design.

Types of bias (when sampling or surveying):

- **Sampling bias:** The sampling method systematically excludes or over-represents some groups (e.g. convenience sample of one class).
- **Undercoverage:** Some groups in the population are not in the sampling frame or have little chance of being

selected.

- **Nonresponse bias:** People who don't respond differ from those who do.
- **Response bias:** The way questions are asked or the setting leads to systematically wrong answers.
- **Voluntary response bias:** Only people who choose to respond are included; they often differ from the population.

Ways to reduce bias: Use a good sampling frame, avoid convenience/voluntary samples when possible, use random sampling (e.g. SRS, stratified), try to reduce nonresponse and careful questionnaire design to reduce response bias.

Ways to increase precision: Increase sample size; sometimes use stratified or other designs that reduce variability.

Finite population correction (FPC)

So far we assumed the sample is a small fraction of the population, so we used:

- **Proportion:** $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- **Mean:** $SE(\bar{x}) = \frac{s}{\sqrt{n}}$

When we **sample without replacement** from a **finite population** of size N , and n is not negligible compared to N , we multiply the standard error by the **finite population correction (FPC)**:

$$FPC = \sqrt{\frac{N-n}{N-1}}.$$

(When N is large, $\sqrt{\frac{N-n}{N}}$ is a common approximation.)

Interpretation: When we draw a large fraction of the population, we get more information than the “infinite population” formula suggests, so the standard error **shrinks** (the FPC is < 1 when $n > 1$). When n is small relative to N , the FPC is close to 1 and we usually ignore it.

95% confidence interval for a proportion (with FPC):

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \cdot \sqrt{\frac{N-n}{N-1}}.$$

95% confidence interval for a mean (with FPC):

$$\bar{x} \pm 2\frac{s}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}.$$

When to use: When the population is finite and n/N is large (e.g. n is more than about 5–10% of N), the FPC narrows the interval. When n/N is small, the FPC is near 1 and the usual formulas are fine.

Example: Proportion with and without FPC

Setting: A small clinic has $N = 50$ patients who had a certain procedure. We take a random sample of $n = 20$ and find $x = 8$ had a positive outcome. So $\hat{p} = 8/20 = 0.40$.

Without FPC (infinite-population formula):

- $SE(\hat{p}) = \sqrt{\frac{0.40 \times 0.60}{20}} = \sqrt{0.012} \approx 0.1096$.
- Margin $\approx 2(0.1096) = 0.219$. **95% CI:** $0.40 \pm 0.219 \rightarrow$ about **[0.18, 0.62]**.

With FPC:

- $FPC = \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{50-20}{49}} = \sqrt{\frac{30}{49}} \approx 0.782$.
- $SE(\hat{p}) \times FPC \approx 0.1096 \times 0.782 \approx 0.0857$. Margin $\approx 2(0.0857) = 0.171$.

- **95% CI (with FPC):** $0.40 \pm 0.171 \rightarrow$ about **[0.23, 0.57]**.

The FPC narrows the interval because we sampled a large fraction ($20/50 = 40\%$) of the population, so we have more information than the infinite-population formula assumes.

Example: Mean with and without FPC

Setting: A small population of $N = 30$ plots. We randomly select $n = 10$ and measure a variable (e.g. plant count per plot). The data are: **4, 5, 6, 7, 7, 8, 9, 10, 11, 12**.

- **Sample mean:** $\bar{x} = (4 + 5 + 6 + 7 + 7 + 8 + 9 + 10 + 11 + 12)/10 = 79/10 = 7.9$.
- **Sample SD:** Deviations from 7.9: $-3.9, -2.9, -1.9, -0.9, -0.9, 0.1, 1.1, 2.1, 3.1, 4.1$. Squared: 15.21, 8.41, 3.61, 0.81, 0.81, 0.01, 1.21, 4.41, 9.61, 16.81. Sum = 60.9. $s^2 = 60.9/9 \approx 6.77$, so $s \approx 2.60$.
- $SE(\bar{x}) = s/\sqrt{n} = 2.60/\sqrt{10} \approx 0.822$.

Without FPC (infinite-population formula):

- Margin $\approx 2(0.822) = 1.64$. **95% CI:** $7.9 \pm 1.64 \rightarrow$ about **[6.3, 9.5]**.

With FPC:

- $FPC = \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{30-10}{29}} = \sqrt{\frac{20}{29}} \approx 0.830$.
- $SE(\bar{x}) \times FPC \approx 0.822 \times 0.830 \approx 0.682$. Margin $\approx 2(0.682) = 1.36$.
- **95% CI (with FPC):** $7.9 \pm 1.36 \rightarrow$ about **[6.5, 9.3]**.

Again, the FPC gives a slightly narrower interval because we sampled a substantial fraction ($10/30 \approx 33\%$) of the finite population.

Infinite-population formulas (full conditions)

When the population is treated as **infinite** (or n/N is very small), use these versions and conditions:

One proportion (Wald), 95% CI: $\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Conditions:

1. **Binomial random process:** (a) **Binary** outcome, (b) **Independent** trials, (c) **Number** of trials n fixed, (d) **Same** probability p on each trial.
2. **Normal approximation:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.

One mean, 95% CI: $\bar{x} \pm 2\frac{s}{\sqrt{n}}$ (or use t for small n).

Conditions:

1. **Either** the population is (approximately) **normal**, **or** $n > 30$ (so the sampling distribution of $\bar{x}$ is approximately normal by the CLT).
2. The sample is a random sample from the population (or can be treated as such).

When the population is **finite** and n/N is not small, multiply the margin of error by the **FPC** $\sqrt{(N-n)/(N-1)}$ for both the proportion and the mean.